Literature Review: Gender Classification Using Facial Recognition

Facial image-based gender classification has become a widely researched topic in biometric security, driven by its applications in surveillance, access control, demographic analytics, and human-computer interaction. Unlike traditional biometric traits such as fingerprints or iris scans, facial imagery allows passive, non-contact acquisition, making it highly suitable for public and real-time systems. This literature review evaluates both traditional and modern approaches for automated gender classification, with a focus on performance, computational efficiency, and practical deployment constraints.

Several works have investigated classical techniques such as **Principal Component Analysis (PCA)** and **Linear Discriminant Analysis (LDA)** for dimensionality reduction, followed by simple classifiers like **k-Nearest Neighbors (k-NN)**. In one study, PCA and LDA were combined to extract low-dimensional features from FERET dataset images, and gender classification was achieved with high interpretability and moderate accuracy [1]. Another comparative analysis found that k-NN consistently performed well across multiple conditions and outperformed more complex models in low-resource settings [2]. Additional work explored edge-based feature extraction combined with neural networks, demonstrating that even with very limited training data, high classification accuracy could be achieved by relying on geometric facial landmarks [3].

More advanced feature engineering methods have also emerged, such as **Multiscale Facial Fusion** using **Local Binary Patterns (LBP)** and **Local Phase Quantization (LPQ)**. This approach, which incorporated multi-block and multi-level descriptors, significantly improved classification performance on unconstrained datasets like Images of Groups (IoG), outperforming several baseline methods [4]. Similarly, PCA and LBP features were used in another study for lightweight facial recognition and gender detection under varying lighting conditions, proving effective in real-world deployments like attendance monitoring systems [9].

While classical approaches remain relevant in constrained environments, **deep learning models** have gained prominence for their accuracy in complex conditions. A VGGNet-based CNN architecture was proposed for gender classification using social media-style face images, achieving robust performance even with limited training samples [5]. Another study evaluated multiple deep learning models, including GoogLeNet and AlexNet, and reported improved accuracy on large datasets like IMDB and FER-2013 [6]. However, these systems are often resource-intensive and require substantial data for training, making them less ideal for embedded or low-power biometric applications.

Several papers also highlight the **limitations** of current systems. For instance, real-time CNN models deployed in human-robot interaction scenarios suffer from decreased performance under occlusion or varied lighting [8]. Other works point to reduced accuracy when scaling to larger k-values in k-NN or dealing with unbalanced gender distributions in datasets [9, 10]. A 3D face recognition study introduced the **Radon transform** in conjunction with PCA and LDA, demonstrating improved resilience to pose variation and shadows, but also noted increased computational demands [10].

Across the literature, a wide range of datasets have been used, including FERET, IoG, UTKFace, and CASIA-3D, each varying in size, pose diversity, and real-world resemblance. Accuracy typically ranged from 85% to 98%, with deep learning methods outperforming traditional ones in unconstrained scenarios, though often at higher computational costs.

In summary, the reviewed literature reveals a clear progression from traditional handcrafted techniques to more data-driven deep learning approaches in gender classification. While classical methods such as PCA, LDA, and k-NN remain effective in constrained or small-sample environments, modern approaches employing CNNs and multiscale fusion features have demonstrated superior performance in real-world, unconstrained scenarios. The inclusion of texture-based features like LBP and LPQ, as well as 3D feature enhancement through techniques like the Radon transform, further expands the classification toolkit. Across all systems, model accuracy remains highly dataset-dependent, with deeper models requiring more data and computational resources. Limitations such as lighting sensitivity, class imbalance, and variability in pose or occlusion are still prevalent, prompting ongoing research into hybrid and robust feature extraction strategies. Overall, the field continues to balance accuracy with efficiency, robustness, and fairness in applied biometric contexts.

References

[1] D. Narmada and D. D. Sudhir, *Gender classification using face recognition*, Int. J. Emerg. Trends Technol. Comput. Sci., vol. 3, no. 4, pp. 214–217, 2014.

[2] M. R. Aqeel et al., *Comparison of several classification algorithms for gender recognition from face images*, Int. J. Adv. Comput. Sci. Appl., vol. 10, no. 8, pp. 331–335, 2019.

[3] K. Ramesha, K. B. Raja, and L. M. Patnaik, *Feature extraction based face recognition, gender and age classification*, Int. J. Comput. Appl., vol. 1, no. 22, pp. 81–86, 2010.

[4] H. Zhang, W. Xu, C. Lyu, and L. Zhao, *Gender classification based on multiscale facial fusion feature*, Math. Probl. Eng., vol. 2018, pp. 1–9, 2018.

[5] T. Ismail and S. A. Talib, *Gender recognition through face using deep learning*, Int. J. Eng. Res. Technol., vol. 7, no. 5, pp. 372–375, 2018.

[6] N. A. Ibraheem, S. O. T. Zubair, and M. A. Elsisi, *Face-based age and gender classification using deep learning model*, Procedia Comput. Sci., vol. 180, pp. 80–85, 2021.

[7] R. S. Pradhan et al., *A comparative study on machine learning techniques for gender classification using facial images*, Int. J. Eng. Trends Comput. Sci., vol. 6, no. 5, pp. 28–35, 2019.

[8] A. Yar, T. Jan, A. Hussain, and U. Din, *Real-time facial emotion recognition and gender classification for human-robot interaction using CNN*, Int. J. Adv. Comput. Sci. Appl., vol. 11, no. 7, pp. 639–645, 2020.

[9] T. V. Swamy and S. G. Ganesh, *Enhancing facial recognition accuracy through KNN conducting*, Int. J. Eng. Explor. Res., vol. 10, no. 3, pp. 63–65, 2022.

[10] M. T. Choudhary and P. S. Zambare, *3D face recognition based on Radon transform, PCA, LDA using KNN and SVM*, Int. J. Sci. Res. Comput. Sci. Eng. Inf. Technol., vol. 3, no. 6, pp. 117–122, 2018.